## Sinusoidal waves

amplitude, angular frequency

$$y = A \sin(kx - \omega t + \phi)$$

angular wave number, phase constant

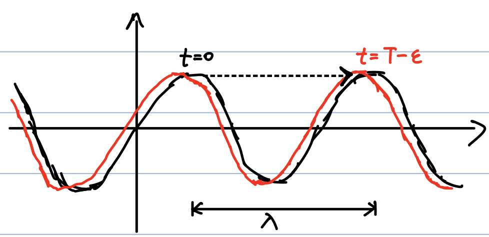
$$y = A \sin(kx - \omega t + \phi)$$

$$x - vt \Rightarrow \quad v = \frac{\omega}{k}$$

$$k = \frac{2\pi}{\lambda}, \quad \omega = \frac{2\pi}{T}. \quad v = \frac{\omega}{k} = \frac{\lambda}{T}$$

$$\Rightarrow \quad y = A \sin\left[\frac{2\pi}{\lambda}(x - vt) + \phi\right]$$

More on speed $v$:




$$y(x, t=T) = A \sin\left[2\pi\frac{x}{\lambda} - 2\pi + \phi\right]$$

$$= y(x, t=0)$$

$$v = \frac{\lambda}{T}$$

Velocity & Acceleration :

$$\begin{cases} v_y = \dfrac{\partial y}{\partial t} = -\omega A \cos(kx - \omega t + \phi) \\ a_y = \dfrac{\partial^2 y}{\partial t^2} = -\omega^2 A \sin(kx - \omega t + \phi) \end{cases}$$

Rate of energy transfer.

Kinetic energy:

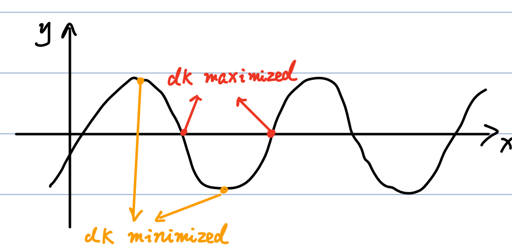
$$dK = \frac{1}{2} dm\, v_y^2$$

$$= \frac{1}{2}(\mu dx)\ \omega^2 A^2 \cos^2(kx - \omega t + \phi)$$

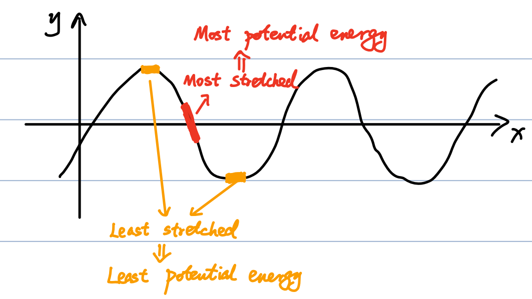
i.e. Kinetic energy travels along with the wave

Note :

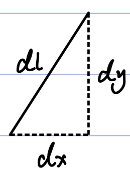
$$\begin{cases} y \sim \sin(kx - \omega t + \phi) \\ dK \sim \cos^2(kx - \omega t + \phi) \end{cases}$$



Potential energy




SAME energy distribution for kinetic and potential !!!




$$dU = T\,(dl - dx)$$

$$= T\left(\sqrt{dx^2 + dy^2} - dx\right)$$

$$\approx \frac{1}{2} T\, dx \left(\frac{dy}{dx}\right)^2$$

$$= \frac{1}{2} T\, k^2 A^2\, dx \, \cos^2(kx - wt + \phi)$$

$$v = \sqrt{T/u} \Rightarrow dU = \frac{1}{2} u\, v^2 k^2 A^2\, dx\, \cos^2(kx - wt + \phi)$$ $v = w/k$

$$= \frac{1}{2} u\, w^2 A^2\, dx\, \cos^2(kx - wt + \phi)$$

$$= dK$$ !

energy density

$$u = \frac{dK + dU}{dx} = u\, w^2 A^2 \cos^2(kx - wt + \phi)$$

$\Rightarrow$ average energy density:

$$u_{avg} = \frac{1}{2} u\, w^2 A^2$$

Note: $\int_0^T \cos^2(wt)\, dt \Big/ \int_0^T dt = \frac{1}{2}$

$\Rightarrow$ average power from source:

$$P_{avg} = u_{avg} \cdot \frac{\lambda}{T}$$

$$= \frac{1}{2} u\, v\, w^2 A^2$$

Interference.

$$\begin{cases} y_1 = A \sin(kx - wt) \\ y_2 = A \sin(kx - wt + \phi) \end{cases}$$

Principles of superposition

$$\Rightarrow \quad y = y_1 + y_2$$

$$= 2A \sin\left[\frac{(kx-\omega t) + (kx-\omega t+\phi)}{2}\right] \cos\left[\frac{(kx-\omega t) - (kx-\omega t+\phi)}{2}\right]$$

$$= 2A \cos\frac{\phi}{2} \sin\left(kx-\omega t+\frac{\phi}{2}\right)$$

1. Constructive interference (in phase)

$$\cos\frac{\phi}{2} = \pm 1 \Rightarrow y = \pm 2A \sin\left(kx-\omega t+\frac{\phi}{2}\right)$$

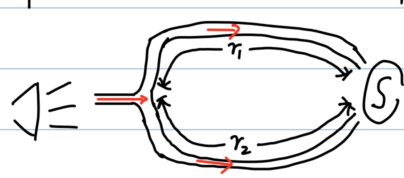
$(\phi = 2\pi n)$

2. Destructive interference (out of phase)

$$\cos\frac{\phi}{2} = 0 \Rightarrow y = 0$$

$(\phi = 2\pi(n+\frac{1}{2}))$

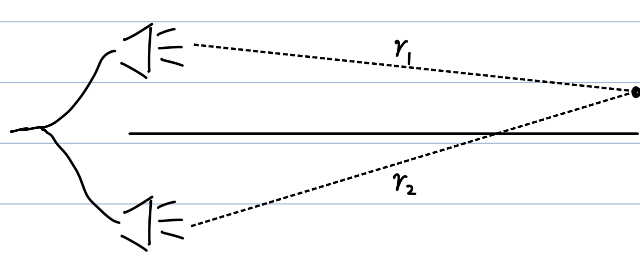
Example. sound cancellation (different $x$ instead of $\phi$)




$\Delta r = |r_1 - r_2| = (n+\frac{1}{2})\lambda$ out of phase (destructive)

$$\Rightarrow k\Delta r = \frac{2\pi}{\lambda}(n+\tfrac{1}{2})\lambda = 2\pi(n+\tfrac{1}{2})$$

Example. Two speakers from same source




$$\Delta r = |r_1 - r_2| = \begin{cases} n\lambda & \text{constructive} \\ (n+\frac{1}{2})\lambda & \text{destructive} \end{cases}$$

Temporal interference

( previous: $w_1 = w_2 = w \Rightarrow y_{total}$ has angular frequency $w$ )

$$\begin{cases} y_1 = A \sin ( k_1 x - w_1 t + \phi_1 ) \\ y_2 = A \sin ( k_2 x - w_2 t + \phi_2 ) \end{cases}$$

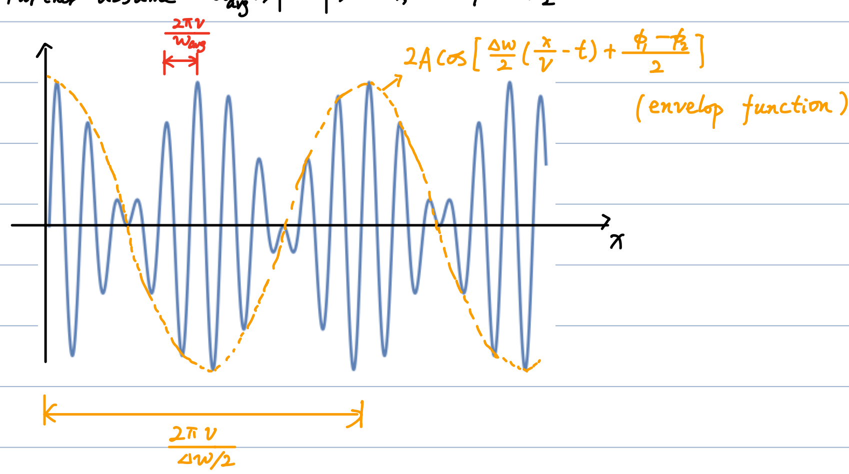
① Assume $v_1 = v_2 = v$ . i.e. $\frac{w_1}{k_1} = \frac{w_2}{k_2} = v$

$$\begin{aligned} y &= y_1 + y_2 \\ &= A \sin \left[ w_1 \left( \frac{x}{v} - t \right) + \phi_1 \right] + A \sin \left[ w_2 \left( \frac{x}{v} - t \right) + \phi_2 \right] \\ &= 2A \sin \left[ \frac{w_1 + w_2}{2} \left( \frac{x}{v} - t \right) + \frac{\phi_1 + \phi_2}{2} \right] \cos \left[ \frac{w_1 - w_2}{2} \left( \frac{x}{v} - t \right) + \frac{\phi_1 - \phi_2}{2} \right] \end{aligned}$$

denote $\begin{cases} \frac{w_1 + w_2}{2} = w_{avg} \\ w_1 - w_2 = \Delta w \end{cases}$

$$\Rightarrow y = \left\{ 2A \cos \left[ \frac{\Delta w}{2} \left( \frac{x}{v} - t \right) + \frac{\phi_1 - \phi_2}{2} \right] \right\} \sin \left[ w_{avg} \left( \frac{x}{v} - t \right) + \frac{\phi_1 + \phi_2}{2} \right]$$

Further assume $w_{avg} \gg |\Delta w|$ , i.e., $w_1 \approx w_2$




Notes:

1. For short distance, we still have wave length

$$\lambda \sim \frac{2\pi}{k} \sim \frac{2\pi v}{w_{avg}}$$

2. For long distance, modulated with a wavelength

$$\lambda' \sim \frac{2\pi v}{\Delta w / 2}$$

3. Such phenomenon : Beat (拍)

$f_{beat} = \frac{1}{2\pi} | w_1 - w_2 |$ : Beat frequency

4. Useful : given $w_1$ , can be used to tune $w_2$ until $w_2 = w_1$

② No longer assume $v_1 = v_2$

$$y = y_1 + y_2$$

$$= A \sin(k_1 x - w_1 t + \phi_1) + A \sin(k_2 x - w_2 t + \phi_2)$$

$$= 2A \cos\left(\frac{k_1 - k_2}{2} x - \frac{w_1 - w_2}{2} t + \frac{\phi_1 - \phi_2}{2}\right) \sin\left(\frac{k_1 + k_2}{2} x - \frac{w_1 + w_2}{2} t + \frac{\phi_1 + \phi_2}{2}\right)$$

$$= 2A \cos\left(\frac{\Delta k}{2} x - \frac{\Delta w}{2} t + \frac{\Delta \phi}{2}\right) \sin\left(k_{avg} x - w_{avg} t + \phi_{avg}\right)$$

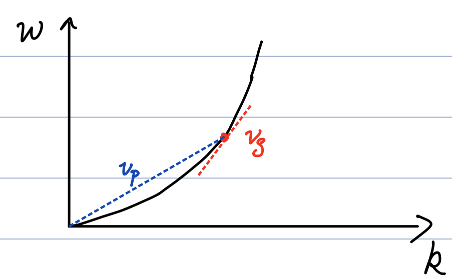
Group velocity: $v_g = \dfrac{\Delta w}{\Delta k}$ (velocity of the envelop function)

for $\Delta w \to 0$, $\Delta k \to 0$

$$v_g = \frac{dw}{dk}$$

Phase velocity: $v_p = \dfrac{w}{k}$

Dispersion relation (for a series of waves):




$v_g = v_p$ : non-dispersive

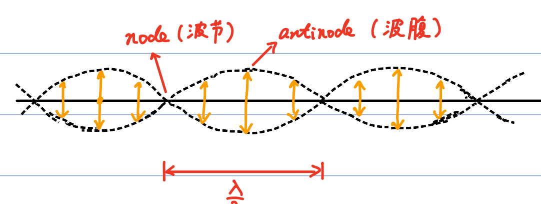
$v_g < v_p$ : normal dispersion

$v_g > v_p$ : anomalous dispersion

Standing waves

$$\begin{cases} y_1 = A \sin(kx - wt) \\ y_2 = A \sin(kx + wt) \end{cases}$$ opposite propagation direction

$$\Rightarrow \quad y = y_1 + y_2 = 2A \sin(kx) \cos(wt)$$

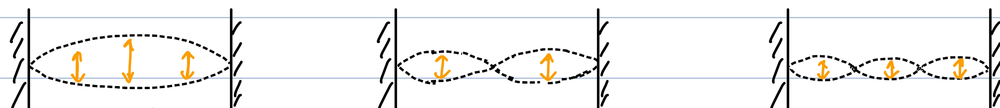



nodes: $kx = n\pi \quad \Rightarrow \quad x = \frac{n}{2}\lambda$

antinodes: $kx = (n + \frac{1}{2})\pi \quad \Rightarrow \quad x = (\frac{n}{2} + \frac{1}{4})\lambda$

- Just oscillation, No propogation of wave
- Each point oscillates with amplitude $2A\sin(kx)$
- No energy transmitted across node

String fixed at both ends :



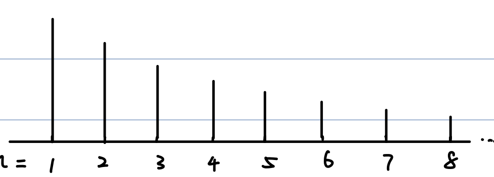
$L = \frac{\lambda_1}{2}$ $\qquad$ $L = \lambda_2$ $\qquad$ $L = \frac{3}{2}\lambda_3$

Generally,

$$\lambda_n = \frac{2L}{n} \qquad \left( v = \frac{\omega}{k} \Rightarrow v \cdot \frac{2\pi}{\lambda} = 2\pi f \right)$$

$$f_n = \frac{v}{\lambda_n} = n\frac{v}{2L} \qquad \text{Harmonic series}$$

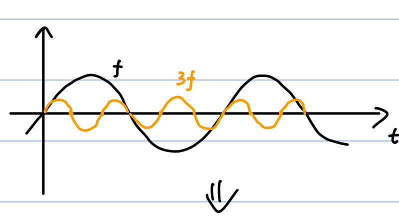
Sounds in music:

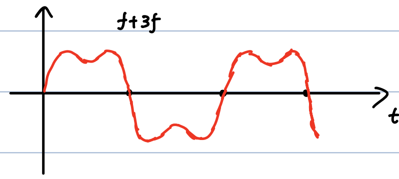



- pitch (frequency) 音高
- loudness (amplitude) 响度
- timbre (tone quality) 音色

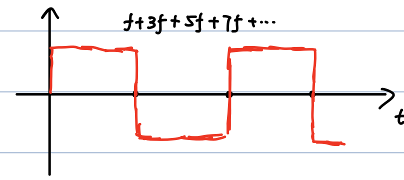
Fourier Analysis

(applies to both space and time, use time as example below)






$\Rightarrow$




- A square wave of period $T$ constructed from $T, \frac{T}{3}, \frac{T}{5}, \cdots$
- Any periodic function $f(t)$ with period $T = \frac{2\pi}{w}$,

$$f(t) = f(t+T)$$

can be expanded in Fourier series:

$$f(t) = \sum_{n=0}^{\infty} \left[ A_n \sin\left(n \frac{2\pi}{T} t\right) + B_n \cos\left(n \frac{2\pi}{T} t\right) \right]$$

$$= B_0 + \sum_{n=1}^{\infty} A_n \sin(nwt) + \sum_{n=1}^{\infty} B_n \cos(nwt)$$

How to calculate $A_n$, $B_n$?

For $m \geq 1$, $n \geq 1$:

$$\begin{cases} \frac{2}{T} \int_0^T \sin(mwt) \sin(nwt)\, dt = \delta_{m,n} \\ \frac{2}{T} \int_0^T \cos(mwt) \cos(nwt)\, dt = \delta_{m,n} \end{cases}$$

$$\Rightarrow \begin{cases} B_0 = \frac{1}{T} \int_0^T f(t)\, dt \\ A_m = \frac{2}{T} \int_0^T f(t) \sin(mwt)\, dt \qquad m \geq 1 \\ B_m = \frac{2}{T} \int_0^T f(t) \cos(mwt)\, dt \qquad m \geq 1 \end{cases}$$

Aperiodic function (e.g. pulse)

$$f(t) = \int_{-\infty}^{\infty} dw\, e^{iwt} F(w)$$

$$\Rightarrow F(w) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt\, e^{-iwt} f(t)$$

Note: $\int_{-\infty}^{\infty} e^{ixy} \frac{dy}{2\pi} = \delta(x)$